

19 | OVERVIEW


Protecting Health in a Warming World: How C-EARTH Tests Local Solutions

Abstract

Extreme heat harms health directly and threatens food and water supplies, especially in communities that have experienced disinvestment. The Center for Effective and Accessible Research-based Testing for Health (C-EARTH) tests practical ways to protect people from heat in Boston, South Africa, and Madagascar. Each solution is chosen with local partners to fit the community's most pressing needs.

Introduction

Climate change is causing more extreme temperatures. Some communities are hit harder than others because residents lack the means or infrastructure to protect themselves, or because of discrimination and past disinvestment. What keeps people safe from heat in one place may not work in another. An air conditioner might protect an older adult in a Boston apartment, while a fishing village without refrigeration faces a different danger: spoiled food. C-EARTH, a center funded by the U.S. National Institutes of Health, works across three very different settings to find heat solutions that are achievable, sustainable, and suited to each community.

How Do Scientists Study This?

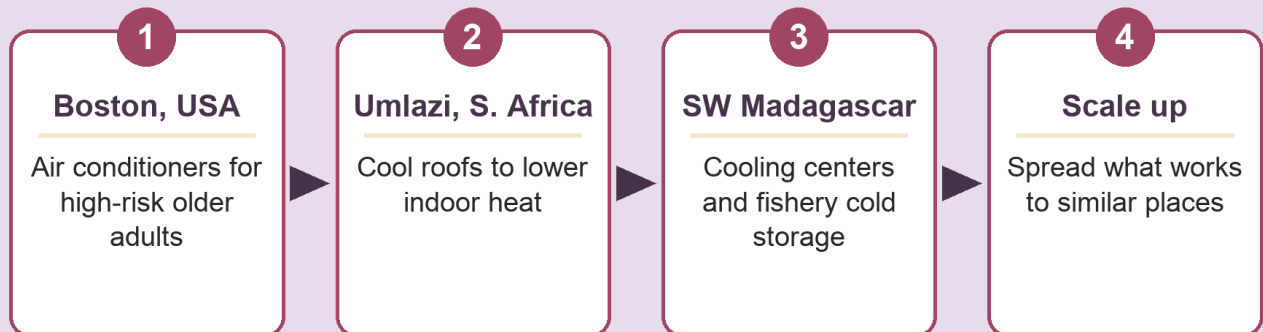


FROM THE ARTICLE

C-EARTH research locations: Boston, South Africa, and Madagascar. Map: Harvard Chan C-CHANGE.

C-EARTH researchers work closely with stakeholder groups in each community to identify their most pressing challenges, then test solutions together. In Boston, they focus on Roxbury, East Boston, Mattapan, and Dorchester, partnering with The Family Van, a free mobile clinic, and Communities Responding to Extreme Weather (CREW). In Umlazi, South Africa, they work with the ARISE Network of 21 institutions from nine sub-Saharan African countries. In 12 coastal villages in southwest Madagascar, they partner with Reef Doctor and the National Network of Women in Fisheries. In each place, researchers evaluate tailored interventions to see whether they reduce heat stress and protect health.

Three places, three solutions



Study the Process

Identify needs

Local partners help researchers understand each community's most pressing heat challenges.

Tailor solutions

Interventions are matched to each setting, from air conditioners to cool roofs to cold storage.

Test and evaluate

Researchers measure whether each solution reduces heat stress and improves health.

What Did They Discover?

C-EARTH is testing three tailored interventions. In Boston, it evaluates air conditioners for high-risk older adults. In South Africa, it tests cool roofs, which reflect sunlight to keep homes cooler. In Madagascar, it studies community cooling centers and cold storage units for fisheries, which can reduce foodborne illness when heat spoils seafood. The Madagascar work also includes building artificial coral reef patches to rebuild fisheries and improve access to seafood amid famine-like conditions. Because solutions are designed with communities, they are more likely to be adopted. If effective, these strategies could be scaled up to save lives in these communities and others facing similar challenges.

C-EARTH by the numbers

3

countries: United States, South Africa, Madagascar

12

coastal villages in southwest Madagascar

21

institutions in the ARISE Network partnership

The work faces real-world obstacles: a 2025 federal funding freeze interrupted a plan to distribute 100 air conditioners to Boston residents.

19 | TAKEAWAYS

Takeaways

Heat is unequal

Communities without resources or infrastructure face the greatest heat risks.

One size doesn't fit all

Air conditioners, cool roofs, and cold storage each fit different communities.

Partners make it work

Local organizations help choose solutions that are practical and sustainable.

Key Terms

Heat stress

Strain on the body when it cannot cool itself enough, which can lead to illness or death.

Cool roof

A roof made with reflective materials that bounce sunlight away, keeping buildings cooler.

Cooling center

A public air-conditioned space where people can escape dangerous heat.

Foodborne illness

Sickness caused by eating contaminated or spoiled food, a greater risk when heat breaks the cold chain.

Disinvestment

A long-term withdrawal of money and resources from a community, often leaving it less able to cope with hazards.

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Activity

Match the Solution

Students match heat solutions to communities and justify their choices based on local needs.

Materials: Community profile cards for Boston, Umlazi, and a Madagascar fishing village;
Solution cards: air conditioner, cool roof, cooling center, cold storage; Chart paper

1. Read each community profile and underline the biggest heat-related risk.
2. Match each solution card to the community where it would help most. Explain your reasoning.
3. For one match, list one challenge that could make the solution hard to sustain, such as electricity costs.
4. Suggest a local partner that could help overcome that challenge.
5. Discuss: why do researchers test solutions in different places instead of choosing one for everyone?

References

Harvard T.H. Chan School of Public Health, C-CHANGE. Center for Effective and Accessible Research-based Testing for Health (C-EARTH).

<https://hsph.harvard.edu/research/climate-health-c-change/research/c-earth/>